SOFTWARE REQUIREMENTS SPECIFICATION

Development of a Multimodal Vision-Language Framework for Human Activity Recognition and Real-Time Safety Event Detection

Hardware-Constrained, Real-Time MVP Specification

Document Item	Value
Document Type	Software Requirements Specification (SRS)
Version	1.0
Status	Baseline / Implementation SRS
Target Hardware	Lenovo LOQ Gen 12-class laptop; RTX 3050-class NVIDIA GPU; 16 GB RAM; 8-core-class CPU; 100+ GB available SSD
Primary Deployment	Local laptop, with optional event-triggered external VLM
Primary UI	React web dashboard
Backend	Python + FastAPI + WebSocket
Database	PostgreSQL (optional during early prototype)

Hardware note: Exact RTX 3050 VRAM varies by laptop configuration. The system shall be designed for a constrained 6 GB-class profile and shall not require continuous execution of large VLMs or heavy video transformers.

Scope decision: The MVP intentionally removes SlowFast fighting detection, Video Swin, TimeSformer, large local Qwen VLM inference, simultaneous heavy video models, and a complex multimodal fusion network. Image/video remain supported inputs.

Revision History

Version	Date	Author/Owner	Change
1.0	September 2026	Project Team	Baseline SRS for hardware-constrained MVP architecture

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1. Introduction

This Software Requirements Specification defines the requirements, architecture, interfaces, data model, machine-learning components, performance targets, testing criteria, and three-month implementation plan for a real-time computer-vision safety monitoring system. The system accepts live or recorded video and may accept still images for event/evidence analysis. It detects people and relevant PPE objects, maintains identities across frames, extracts human pose, recognizes a constrained set of human activities, detects selected safety events, calculates a deterministic risk score, and presents alerts through a web dashboard.

The design is intentionally optimized for a Lenovo LOQ Gen 12-class development laptop with an RTX 3050-class GPU and 16 GB RAM. Because the GPU is the primary resource constraint, the system uses lightweight models in the real-time path and treats vision-language analysis as optional and event-triggered rather than continuously resident on the GPU.

1.1 Purpose

The purpose of the system is to provide explainable, near-real-time safety decision support from visual streams. The system is not intended to replace human safety personnel, emergency response, or certified safety systems.

1.2 Document Conventions

Keyword	Meaning
MUST	Mandatory requirement for the MVP.
SHOULD	Recommended requirement that may be deferred if necessary.
MAY	Optional enhancement.
FR	Functional Requirement identifier.
NFR	Non-Functional Requirement identifier.
MLR	Machine-Learning Requirement identifier.
API	Application Programming Interface.

2. Overall Description

2.1 Product Perspective

The product is a modular AI application consisting of a video ingestion layer, object detection and tracking, pose extraction, lightweight temporal activity recognition, specialized safety-event logic, risk scoring, optional event-triggered vision-language analysis, a FastAPI backend, PostgreSQL persistence, and a React dashboard.

```
VIDEO / IMAGE INPUT -> OpenCV/FFmpeg -> YOLO11n -> ByteTrack -> MediaPipe Pose
Pose/Motion Features -> Small Temporal Transformer -> Activity
Activity + PPE + Zones -> Safety Event Engine -> Risk -> Evidence -> Optional VLM
Explanation -> FastAPI/WebSocket -> React Dashboard
```

2.2 Product Functions

- Ingest live webcam/RTSP-like streams, recorded videos, and still images for selected analysis flows.
- Detect people and relevant objects using a lightweight YOLO model.
- Assign and maintain person identities using ByteTrack.

- Estimate human body landmarks using MediaPipe Pose.
- Recognize a small set of general human activities from temporal pose/motion sequences.
- Detect fall, no-helmet, and restricted-area-entry safety events.
- Calculate deterministic risk scores from event severity, confidence, duration, and context modifiers.
- Generate rule-based explanations in the core system and optionally enrich event explanations with an external or lightweight vision-language model.
- Display live video, detections, activity labels, alerts, risk scores, event history, and explanations.
- Persist event metadata and system settings for later review and evaluation.

2.3 Design Constraints

- The baseline target is an RTX 3050-class laptop GPU with 16 GB system RAM.
- The MVP must avoid continuous local inference with large multimodal language models.
- The MVP must avoid running multiple heavy video models simultaneously.
- The real-time path prioritizes stable throughput and low memory pressure over maximum benchmark accuracy.
- The system shall not depend on a cloud service for the basic detection, activity, safety-event, and dashboard workflow.
- The core system must continue to operate if optional VLM enrichment is unavailable.

3. System Scope and Boundaries

In Scope for MVP	Out of Scope / Future
Live/recorded video ingestion	Violence/fighting detection with SlowFast
Still-image evidence analysis	Video Swin or TimeSformer live inference
Person detection + tracking	Large local Qwen-class VLM continuously loaded
MediaPipe Pose	Complex end-to-end multimodal fusion network
Walking, running, standing, sitting, climbing, optional lifting/carrying	Multi-camera distributed fleet management
Fall detection	Autonomous safety control/actuation
No-helmet detection	Automatic emergency dispatch
Restricted-zone detection	High-availability production deployment
Risk scoring and explanations	Medical diagnosis or security adjudication
React dashboard + FastAPI + WebSocket	Biometric identification / face recognition

4. Stakeholders and User Classes

User Class	Needs / Responsibilities
System Administrator	Configures cameras, zones, thresholds, models, and retention settings.
Safety Supervisor	Monitors live alerts, reviews events, acknowledges alerts, and checks explanations.
Research/Development User	Runs experiments, evaluates models, inspects logs, and tunes thresholds.
Project Reviewer / Faculty	Reviews architecture, metrics, demonstrations, and documentation.

5. Functional Requirements

ID	Function	Requirement
FR-01	Video Input	The system shall accept a local video file and a live camera source as primary inputs.
FR-02	Image Input	The system shall accept still images for one-shot analysis and VLM/evidence workflows where enabled.
FR-03	Frame Sampling	The system shall support configurable processing FPS independent of source FPS.
FR-04	Person Detection	The system shall detect people and output bounding boxes and confidence scores.
FR-05	Object/PPE Detection	The system shall detect at least helmet and person classes for PPE compliance.
FR-06	Tracking	The system shall assign stable track IDs using ByteTrack or an equivalent lightweight tracker.
FR-07	Pose Estimation	The system shall extract human body landmarks for tracked persons where pose inference is available.
FR-08	Activity Recognition	The system shall classify at least walking, running, standing, sitting, and climbing from temporal features.
FR-09	Fall Detection	The system shall detect a fall using pose, body-motion, and temporal persistence features.
FR-10	No-Helmet Detection	The system shall associate helmet detections with person tracks and flag missing helmet states according to configured rules.
FR-11	Restricted Zone	The system shall support polygonal restricted zones and flag persistent person entry.
FR-12	Temporal Persistence	The system shall avoid issuing an event solely from one unstable frame when a persistence threshold is configured.
FR-13	Risk Score	The system shall compute a normalized risk score and map it to a severity band.
FR-14	Evidence Capture	The system shall capture a configurable set of pre-trigger/trigger/post-trigger frames for each confirmed event.
FR-15	Explanation	The system shall generate a concise explanation using deterministic event evidence and optionally a VLM.
FR-16	Alerts	The system shall display active alerts and allow acknowledgment.
FR-17	Real-Time Updates	The backend shall push live state updates through WebSocket channels.
FR-18	History	The system shall store and display event history with timestamps, severity, confidence, and explanation.
FR-19	Configuration	The system shall allow camera, zone, threshold, and model settings to be modified without changing source code.
FR-20	Degraded Mode	The system shall continue operating

		without VLM enrichment and shall report optional-service unavailability.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Target 10-15 FPS for the main detection/streaming path at 640-class inference resolution on the target laptop; exact throughput shall be measured and documented.
NFR-02	Latency	For rule-based events, target alert generation within approximately 0.5-1.5 seconds of confirmed trigger under a controlled single-camera test.
NFR-03	Stability	The application shall run a minimum 30-minute continuous test without crash, unbounded memory growth, or persistent GPU OOM.
NFR-04	Resource Control	The system shall expose configurable FPS, frame skipping, queue size, and model-loading controls.
NFR-05	Modularity	Detection, pose, activity, safety rules, VLM, database, and UI shall be separable modules.
NFR-06	Fault Tolerance	Optional VLM failure shall not stop the core safety pipeline.
NFR-07	Usability	A supervisor shall be able to identify an active high-risk event within one dashboard view.
NFR-08	Maintainability	Configuration shall be externalized in YAML/ENV configuration files.
NFR-09	Testability	Core rule engine and API modules shall have automated unit/integration tests.
NFR-10	Privacy	The system shall avoid face recognition or identity inference in the MVP and shall support configurable evidence retention.
NFR-11	Explainability	Every safety alert shall retain machine-readable evidence used to generate the explanation.
NFR-12	Reproducibility	Model names, configuration versions, and threshold settings shall be stored with evaluation results.

7. System Architecture

7.1 Approved Hardware-Constrained Architecture

```
INPUT: Live Video / Recorded Video / Image
        |
   OpenCV / FFmpeg
        |
```



7.2 Component Responsibilities

Component	Selected Technology	Primary Responsibility
Video Ingestion	OpenCV + FFmpeg	Read frames, handle files/cameras, sample FPS, manage stream lifecycle.
Human/Object Detection	YOLO11n	Detect person and selected objects/PPE.
Tracking	ByteTrack	Maintain identities across frames.
Pose	MediaPipe Pose	Extract body landmarks and visibility values.
HAR	Small Temporal Transformer	Classify general activities from short pose/motion sequences.
PPE	Small custom YOLO11n model	Detect helmet/person for PPE compliance.
Fall	Rule/temporal classifier	Detect rapid downward motion + body orientation + persistence.
Restricted Zone	Polygon geometry	Detect persistent entry based on track foot point.
Risk Engine	Python rules	Normalize event severity and contextual modifiers.
VLM	Optional lightweight/local or external event-triggered model	Generate richer natural-language explanation from selected evidence frames.
Backend	FastAPI + WebSocket	Expose APIs and real-time event/state updates.
Database	PostgreSQL	Persist events, detections, configurations, and explanations.
Frontend	React	Render live dashboard, alerts, timeline, event details, settings.

8. Detailed Module Specifications

8.1 Video Ingestion Module

- Input types: webcam, local MP4/AVI/MKV through FFmpeg/OpenCV, and optional network stream where available.
- Configurable source FPS and processing FPS.
- Frame queue shall have a bounded maximum size to prevent memory growth.
- Dropped frames shall not crash the pipeline and shall be reported in diagnostics.
- Image input shall bypass temporal HAR and may directly invoke selected object/VLM analysis workflows.

8.2 Detection and Tracking Module

The baseline detector shall use YOLO11n because the project must preserve GPU headroom for the temporal classifier and UI/backend operations. YOLO11s may be evaluated later if stable throughput remains above the target. Detection shall use 640-class input resolution unless profiling shows a lower resolution is necessary.

Parameter	Baseline
Detector	YOLO11n
Input	640 x 640-class
Precision	FP16 on NVIDIA GPU when supported
Target inference rate	10-15 FPS
Tracker	ByteTrack
Tracked classes	Person
PPE classes	Person, Helmet; optional Vest

8.3 Pose Module

MediaPipe Pose shall operate on person crops or tracked person regions. The module shall return landmark coordinates and visibility/confidence values. Pose data shall be normalized relative to the person bounding box and used to derive temporal features such as joint velocity, body angle, and relative displacement.

8.4 Human Activity Recognition Module

The main activity recognizer shall be a small temporal Transformer operating on a fixed-length sequence of pose and motion features rather than a full RGB video transformer. This choice reduces VRAM usage while retaining explicit temporal modeling.

```
Input per person:
  32 frames x normalized pose/motion features

Example features:
  - normalized x/y/z landmark coordinates
  - landmark velocity
  - selected joint angles
  - body orientation
  - bounding-box width/height
  - center-of-box velocity

Temporal Transformer
  -> activity probability vector
  -> walking / running / standing / sitting / climbing
```

```
-> optional lifting / carrying
```

8.5 Fall Detection Module

Fall detection shall use existing pose/tracking information. A fall candidate shall be generated when a combination of rapid downward movement, change in body orientation, low-body position, and temporal persistence crosses configured thresholds. The detector shall maintain a cooldown period to prevent repeated alerts for one event.

8.6 PPE Module

A separate lightweight YOLO model shall detect helmet and person. Helmet assignment shall use spatial association between the helmet bounding box and the upper region of the corresponding person box. A no-helmet event shall require persistence across multiple processed frames rather than a single missing detection.

8.7 Restricted-Zone Module

Administrators shall draw or configure polygonal restricted zones. For each tracked person, the bottom-center point of the person bounding box shall be tested against the polygon. An event shall be triggered only after configurable persistence, reducing false positives caused by boundary jitter.

8.8 Risk Engine

Risk scoring shall be deterministic and independent of the VLM. A recommended normalized formula is:

```
Risk = 0.35*Severity + 0.25*Confidence + 0.20*ExposureDuration + 0.20*Context

Bands:
 0-29   LOW
 30-59  MEDIUM
 60-79  HIGH
 80-100 CRITICAL
```

Weights and thresholds shall be configurable and must be stored with evaluation metadata. These scores are engineering indicators for prioritization, not certified safety judgments.

8.9 Optional VLM Module

The VLM is not part of the mandatory detection loop. On a confirmed event, the system may capture 3-5 representative evidence frames and structured metadata and send them to a lightweight local model or external vision-language service. The service should return event context, concise explanation, and optional recommendation text. If the VLM fails or is unavailable, the system shall use the deterministic explanation template.

9. Data and Database Requirements

PostgreSQL is the target persistence layer. During the earliest prototype stage, SQLite may be used temporarily to reduce setup overhead, but the final MVP data model shall be compatible with PostgreSQL.

9.1 Entity Model

```
Camera 1---N Detection
Camera 1---N Activity
Camera 1---N Event
Camera 1---N Zone
```

```
Event 1---N Explanation
Event 1---N Alert
Event N---1 Camera
Event N---0..1 Track
```

9.2 Core Tables

Table	Key Fields
cameras	id, name, source_type, source_url, resolution, target_fps, status, created_at
zones	id, camera_id, name, polygon_json, zone_type, risk_modifier, enabled
detections	id, camera_id, timestamp, track_id, class_name, confidence, bbox_x, bbox_y, bbox_w, bbox_h
activities	id, camera_id, track_id, timestamp, activity, confidence, duration_ms
events	id, camera_id, track_id, event_type, severity, risk_score, confidence, started_at, ended_at, status
explanations	id, event_id, model_name, explanation, evidence_json, recommendation, created_at
alerts	id, event_id, alert_type, severity, message, acknowledged, created_at
system_configs	id, key, value_json, version, updated_at

9.3 Data Retention

- Raw video shall not be stored by default unless explicitly enabled for research/evidence use.
- Event metadata may be retained for evaluation and dashboard history.
- Evidence frames shall have configurable retention (default: short-term project use only).
- Logs shall not contain API secrets or private credentials.

10. API Requirements

Method	Endpoint	Purpose
GET	/api/health	System health and component status.
GET	/api/cameras	List configured cameras.
POST	/api/cameras	Create camera configuration.
GET	/api/cameras/{id}	Return camera status/configuration.
DELETE	/api/cameras/{id}	Disable/remove camera configuration.
GET	/api/events	List events with filtering.
GET	/api/events/{id}	Return event detail, evidence and explanation.
POST	/api/events/{id}/acknowledge	Acknowledge an alert/event.
GET	/api/alerts	Return active/recent alerts.
GET	/api/zones	List zones.
POST	/api/zones	Create polygon zone.
PUT	/api/zones/{id}	Update zone.
DELETE	/api/zones/{id}	Delete/disable zone.
GET	/api/settings	Return non-secret configuration.
PUT	/api/settings	Update thresholds and runtime settings.
WS	/ws/camera/{id}	Push live camera state, tracks, activity,

		and alerts.

10.1 WebSocket Payload

```
{
 "timestamp": "ISO-8601",
 "camera_id": 1,
 "fps": 12.4,
 "tracks": [
  {"id": 7, "bbox": [x,y,w,h], "activity": "climbing", "confidence": 0.91}
 ],
 "events": [
  {"type": "NO_HELMET", "severity": "HIGH", "risk_score": 72}
 ]
}
```

11. Dashboard Requirements

11.1 Main Dashboard

```
+-------------------------------------------------------------------+
| SAFETY AI MONITOR                SYSTEM ONLINE      Camera 01   |
+-----------------------------------+-------------------------------+
|                                   | ACTIVE ALERTS                 |
|            LIVE VIDEO             | [HIGH] No Helmet              |
|                                   | [MED ] Restricted Zone        |
|   [Bounding boxes + track IDs]    |                               |
|                                   | CURRENT RISK                  |
| Objects: 8   Persons: 4           |         72 / 100              |
| Activity: Climbing                |                               |
+-----------------------------------+-------------------------------+
| EVENT TIMELINE                    | AI / RULE EXPLANATION         |
| 14:31 No Helmet                   | Worker is climbing...         |
| 14:33 Restricted Zone             | Helmet not detected...        |
+-----------------------------------+-------------------------------+
```

11.2 Required Pages

Page	Required Content
Dashboard	Live video, bounding boxes, track IDs, activity labels, current risk, active alerts, latest explanation.
Events	Search/filterable event list with time, camera, event type, severity, confidence, status.
Event Details	Evidence frames, activity, detections, risk breakdown, explanation, acknowledgment.
Analytics	Event counts, severity distribution, activity distribution, average alert latency, false-alarm review metrics.
Settings	Camera source, processing FPS, safety thresholds, zone polygons, model selection where supported.

12. Dataset and Machine-Learning Requirements

12.1 Dataset Plan

Task	Baseline Dataset Strategy	Purpose
General HAR	UCF101 subset + custom project clips	Train/evaluate basic activities relevant to monitored safety scenarios.
Fall	LE2I / UR Fall / similar permitted research datasets + custom clips	Evaluate fall detection and cross-scene robustness.
PPE	Openly licensed PPE dataset + custom construction-like images	Train helmet/person detector.
Restricted Zone	Synthetic/custom camera scenes	No public dataset required; event is geometry + tracking.
VLM evidence	Custom event snapshots / synthetic prompt-response set	Validate explanation consistency and formatting.

12.2 Model Baseline

Module	Required Model	Rationale
Detection	YOLO11n	Low compute and sufficient baseline for person detection.
Tracking	ByteTrack	Lightweight identity continuity.
Pose	MediaPipe Pose	Efficient live pose extraction.
HAR	Small Temporal Transformer	Uses compact pose/motion sequences; avoids full RGB video transformer.
PPE	Small custom YOLO11n	Specialized detector for helmet/person.
Fall	Pose + temporal logic/classifier	Reuses existing features and avoids another heavy model.
Restricted Zone	Polygon geometry	Negligible compute compared with learned video models.
VLM	Optional event-triggered lightweight/external model	Keeps GPU headroom and preserves core functionality if unavailable.

12.3 ML Evaluation Metrics

- Object/PPE detection: precision, recall, mAP where ground-truth annotations are available.
- Activity recognition: accuracy, macro F1, per-class precision/recall, confusion matrix.
- Fall detection: precision, recall, F1, false alarms per hour, event detection latency.
- Restricted zone: precision, false alarms per hour, entry-to-alert latency.
- System: FPS, median latency, GPU VRAM, system RAM, CPU utilization, dropped frames, crash count.
- VLM: explanation validity, structured-output success rate, hallucination/unsupported-claim review rate.

13. Hardware and Performance Requirements

Resource	Target / Constraint
CPU	8-core-class modern mobile CPU or equivalent.
RAM	16 GB baseline; 32 GB is optional improvement but not required.
GPU	RTX 3050-class NVIDIA laptop GPU; design for constrained VRAM.

Storage	At least 100 GB free for code, datasets, checkpoints, logs and test videos; 200+ GB is preferable for experimentation.
OS	Windows 11 with CUDA-capable NVIDIA stack; Linux/WSL or Ubuntu may be used for reproducible training/deployment.

13.1 Performance Budget

Pipeline Stage	Budget / Policy
Video decode	CPU; bounded queue.
YOLO11n	10-15 FPS target; FP16 if stable.
ByteTrack	CPU; process every available tracking update where practical.
MediaPipe Pose	10-15 FPS target or adaptive sampling.
Temporal HAR	5-10 inference cycles/s depending on active tracks.
PPE detector	Periodic or event-oriented inference; not required to run at maximum FPS.
Risk engine	CPU; event-driven.
VLM	Only after confirmed event; asynchronous where possible.
Dashboard	10-15 UI updates/s target; UI must not block inference thread.

13.2 Stability Requirements

- The system shall not load multiple large neural networks unnecessarily into GPU memory.
- Model inference shall run in worker threads/processes or asynchronous tasks so the UI does not block.
- All queues shall be bounded.
- GPU OOM shall cause graceful degradation or model unload, not full application termination.
- An application watchdog/health endpoint shall report stream, detector, backend, database and optional VLM status.

14. Security, Privacy and Safety Requirements

- API keys and external VLM credentials shall be stored in environment variables or a secure secret store, never in source code.
- The system shall not perform face recognition, identity matching, or biometric identification in the MVP.
- Event evidence shall be stored only as long as necessary for project evaluation or configured retention.
- Dashboard access controls may be added for deployment, but local development may operate without authentication.
- Risk scores and recommendations are decision-support outputs and shall be labeled accordingly.
- The system shall provide visible status when an optional analysis component is unavailable.

15. Testing and Acceptance Criteria

15.1 Unit Tests

- Polygon point-in-zone calculations.

- Helmet-to-person association.
- Fall feature calculations.
- Temporal buffer management.
- Risk score calculation and severity mapping.
- Event cooldown/de-duplication.
- API request validation and serialization.

15.2 Integration Tests

- Video input -> detector -> tracker -> pose -> activity.
- Detector/tracker -> safety event -> event record.
- Event -> evidence capture -> optional VLM -> explanation record.
- Backend -> WebSocket -> React dashboard.
- Database failure -> graceful error reporting.

15.3 Acceptance Criteria

Area	Acceptance Target
Core run	30-minute single-camera run without crash or persistent memory growth.
Real-time	Sustained practical throughput in the 10-15 FPS class on target hardware, measured and documented.
HAR	Demonstrate at least five activity classes with a documented confusion matrix and macro F1.
Fall	Demonstrate event detection on held-out test clips and report false alarms/hour and latency.
Helmet	Demonstrate helmet/no-helmet event behavior under varied person positions.
Restricted zone	Demonstrate zone entry and exit with boundary persistence/debouncing.
Dashboard	Live feed, detections, activity, risk, alerts and event details visible without page refresh.
Degraded mode	Core detection and safety alerts continue if VLM is disabled or unavailable.
Reproducibility	Model/config versions and evaluation commands documented in README.

16. Three-Month Implementation Roadmap

Week	Deliverables	Exit Criteria
1	Repo, environment, GPU/CUDA check, OpenCV/FFmpeg input, FastAPI skeleton	Video stream/file works; health endpoint responds.
2	YOLO11n + ByteTrack	Tracked people with IDs and stable FPS measurement.
3	MediaPipe Pose + feature extraction	Pose overlays and stored temporal feature vectors.
4	Temporal Transformer baseline	5 core activities recognized on test clips.
5	PPE detector + helmet association	Helmet/no-helmet event works on curated test set.
6	Fall detector	Fall event and cooldown

		implemented; initial metrics collected.
7	Restricted-zone engine	Polygon drawing/configuration and entry alerts implemented.
8	Safety event manager + risk engine	Single event model, severity and event de-duplication work.
9	Event evidence capture + optional VLM integration	Event-triggered explanation path works; optional and non-blocking.
10	React dashboard + WebSocket	Live dashboard shows tracks, activity, risk and alerts.
11	Database + history + analytics + settings	Events persist and are reviewable.
12	Profiling, testing, documentation, demo hardening	30-minute stability test passes; final metrics and screenshots captured.

16.1 Milestone Gates

- Milestone 1: Detection + tracking stable before adding temporal HAR.
- Milestone 2: HAR + safety rules stable before adding VLM.
- Milestone 3: Dashboard and database integrated only after core inference loop is stable.
- Final gate: no new major model added during the last evaluation week unless it fixes a measured requirement failure.

17. Risks and Mitigations

Risk	Impact	Mitigation
GPU OOM	High	Use YOLO11n, small temporal model, FP16, avoid simultaneous heavy models, event-trigger VLM.
Low FPS	High	Lower processing FPS, reduce resolution, process pose/PPE periodically, reduce temporal model size.
False fall alerts	High	Temporal persistence, cooldown, multi-feature confirmation, held-out evaluation.
Helmet association errors	Medium	Use upper-person region + IoU/distance association + temporal smoothing.
Boundary jitter at zones	Medium	Use polygon persistence and hysteresis around zone boundary.
VLM unavailable	Medium	Core deterministic explanation fallback; VLM is optional.
Dataset mismatch	High	Add custom clips from target environment and document domain shift.
16 GB RAM pressure	Medium	Bounded queues, avoid storing full videos in memory, close unused model/process resources.
Over-scoping	High	Freeze MVP to three safety events and a small activity set.

18. Future Scope

- Fighting/violence detection using a specialized streaming video model such as SlowFast or MoViNet variants.
- Evaluation of heavier Video Swin or TimeSformer models using a stronger GPU or cloud training environment.
- Larger local VLMs such as Qwen-class 7B models on higher-VRAM hardware.
- Multi-camera synchronization and centralized monitoring.
- Additional PPE classes such as vest, goggles, gloves and boots.
- Fire/smoke detection and equipment proximity warnings.
- Edge deployment on Jetson-class hardware after profiling.
- Human-in-the-loop alert verification and feedback-based threshold tuning.

19. Glossary

Term	Definition
HAR	Human Activity Recognition: classification of a person's activity over time.
VLM	Vision-Language Model: a model that combines visual input with language generation/understanding.
Pose Estimation	Estimation of anatomical landmarks such as shoulders, hips, knees and ankles.
ByteTrack	A lightweight multi-object tracking method that maintains object identities across frames.
Temporal Model	A model that consumes sequences of features/frames and captures changes over time.
PPE	Personal Protective Equipment such as helmets and safety vests.
Risk Score	A deterministic project-specific numerical indicator used to prioritize safety events.
Event Persistence	Minimum duration/number of frames required before an event is confirmed.
Evidence Frame	A selected image from the stream retained to support an event explanation.
Degraded Mode	Operation with optional modules disabled while core functions remain available.

Appendix A - Recommended Configuration Baseline

```
models:
  detector: yolo11n
  tracker: bytetrack
  pose: mediapipe
  activity: temporal_transformer_small
  ppe: yolo11n_ppe
  vlm: optional_event_triggered

runtime:
  input_fps: 30
  detection_fps: 12
  pose_fps: 12
  activity_fps: 6
```

```
  frame_size: 640
  fp16: true
  max_queue: 2
  evidence_frames: 5

safety:
  fall_cooldown_seconds: 10
  zone_persistence_seconds: 1.5
  helmet_persistence_frames: 4
```

Appendix B - Example Event JSON

```
{
  "event_type": "NO_HELMET",
  "camera_id": 1,
  "track_id": 17,
  "activity": "climbing",
  "confidence": 0.93,
  "risk_score": 72,
  "severity": "HIGH",
  "evidence": {
    "helmet_detected": false,
    "person_detected": true,
    "activity_confidence": 0.91
  },
  "explanation": "Worker appears to be climbing while no helmet is detected.",
  "recommendation": "Verify PPE compliance."
}
```

Appendix C - Final MVP Definition

MVP freeze: The approved three-month MVP consists of: video/image input; YOLO11n detection; ByteTrack; MediaPipe Pose; small Temporal Transformer HAR; three safety events (fall, no helmet, restricted zone); deterministic risk engine; optional event-triggered VLM explanation; FastAPI/WebSocket backend; PostgreSQL persistence; React dashboard. Heavy video transformers, continuous large VLM inference, fighting detection, biometric identification, and complex multimodal fusion are excluded from the MVP.